Software Design Document: Engine Control Demand Characterizer

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1 Introduction

1.1 Purpose

This software aims to:

1. Define a simplified engine model (e.g., accel, speed parameters),
2. define a set of Adaptive Variable-Rate (AVR) tasks as proposed by Biondi et al. [?] (e.g., transition speeds, execution times),
3. combine the engine model and AVR task to form a real-time AVR-based engine control model, hereafter AEM (AVR Engine Model),
4. calculate the maximum demand (see definition CITE) a given AEM can generate, and
5. support result synthesis and visualization (e.g. defining experiment classes and parameters, exporting test data into portable files, enabling comparing algorithm runtimes, etc.)

The intended use is demand characterization for generated AEMs. In furtherance of this intended use, the software also aims to enumerate multiple calculation options (e.g., different methods).

1.2 Scope

The scope of this software includes three primary areas:

1. model definition and generation,

2. algorithm implementation,
3. parallelization support, and
4. synthesis and data visualization.

For model definition and generation, scope includes:

1. enumeration of existing models and
2. randomized generation based on existing models.

Excluded items include:

1. creation of new engine models and
2. parameter extension beyond existing models.

For algorithm implementation, scope includes:

1. implementation of existing algorithms and
2. parameterization of algorithm variations where feasible (e.g., parameterization of constants in existing research).

Excluded items include:

1. design and implementation of new algorithms and
2. approximation of algorithms no already specified.

For synthesis and data visualization, scope includes:

1. definition and implementation of experiment templates,
2. parameterization of experiment variables,
3. design and implementation of experiment test harness(es) for exporting key performance indicators (e.g., demand calculated, runtime, RAM use, etc.) , and
4. definition and use of common key performance data export format (e.g., CSV enumeration of key performance indicators).

1.3 Definitions, Acronyms, Abbreviations

- AVR - Adaptive Variable Rate - The Adaptive Variable Rate task model by Biondi et al. CITE
- HPC - High Performance Computing - High performance computing, usually a grid, cluster, or collection of compute devices
- ICE - Internal Combustion Engine - Heat engine using fuel combustion to cause high-temperature, high-pressure gas expansion for performing work
- SI ICE - Spark Ignition ICE - Spark ignition ICE

1.4 Acknowledgements

This work templated from IEEE 830-1984 - IEEE Guide for Software Requirements Specifications [1].

2 General Overview

2.1 Assumptions

2.2 Constraints

2.3 Standards

3 Architecture Design

3.1 Logical View

3.2 Hardware Architecture

3.3 Software Architecture

3.4 Security Architecture

3.5 Communication Architecture

4 System Design

4.1 Use-Cases

4.2 Sequence Diagrams

4.3 Data Flow Diagrams

References

[1] IEEE. IEEE Guide for Software Requirements Specifications. *IEEE Std 830-1984*, pages 1–26, Feb. 1984.